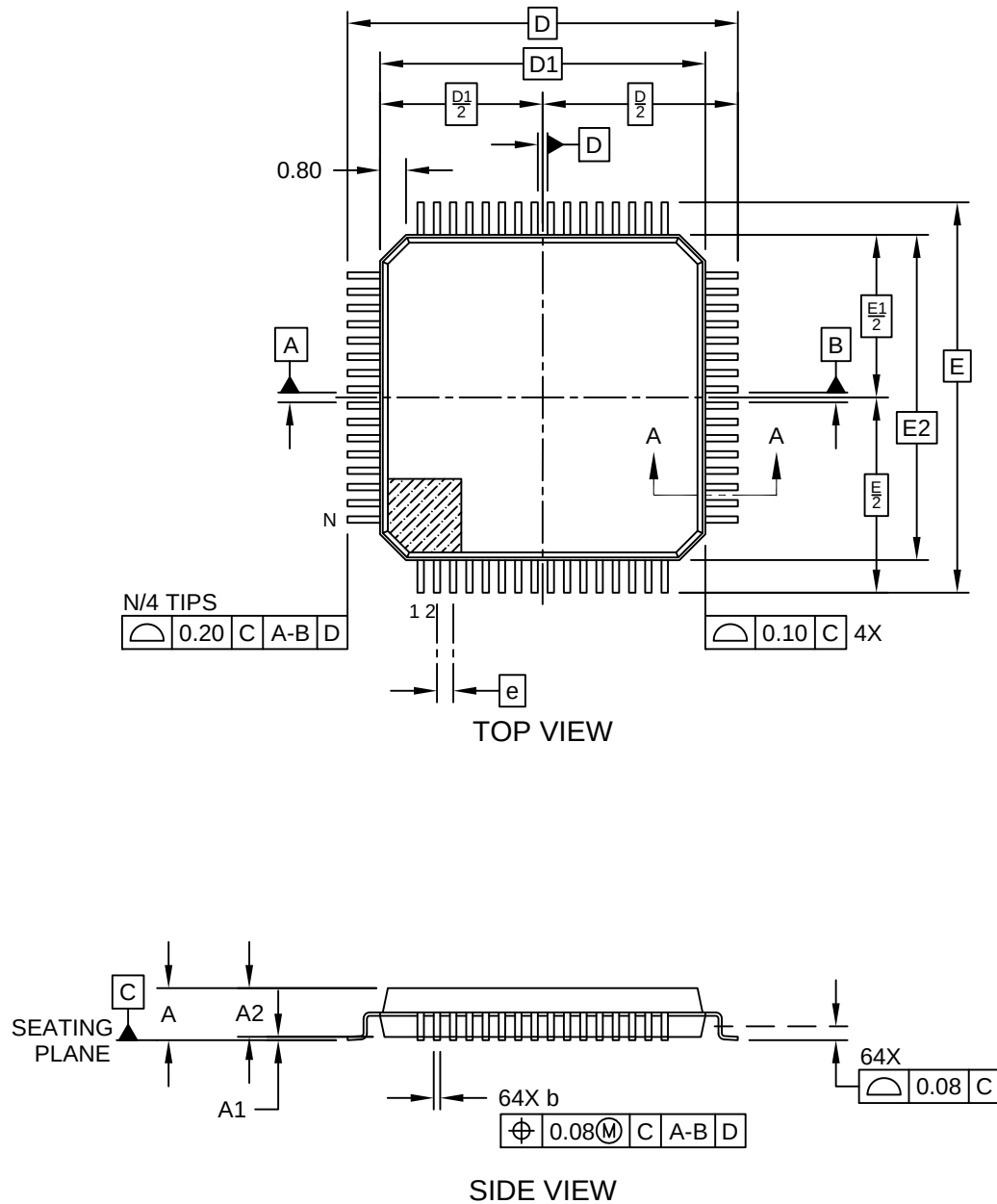


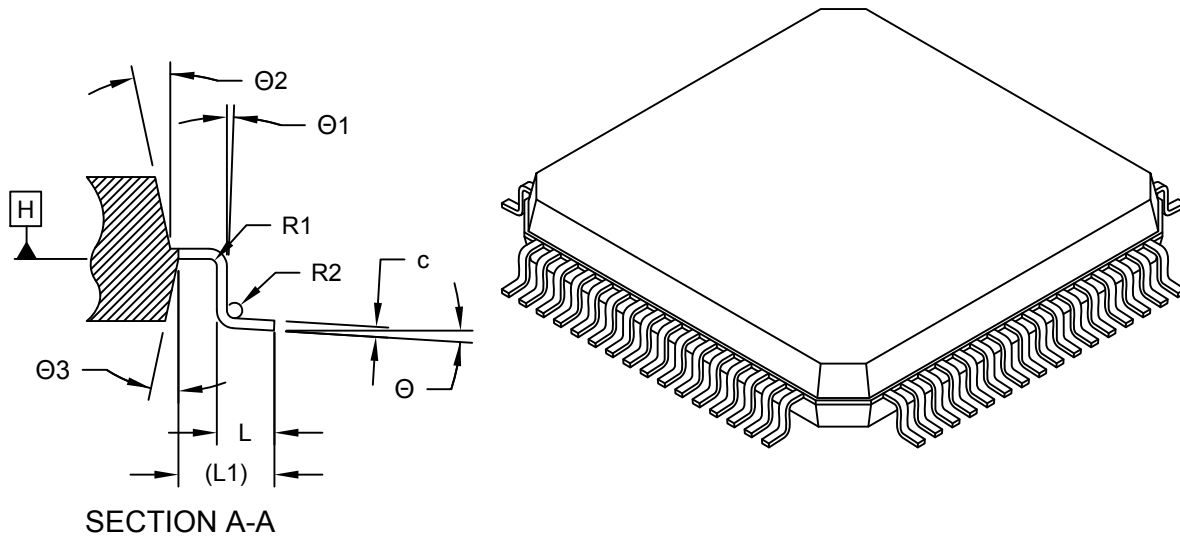
64-Lead Low Profile Quad Flat Pack (2CB) - 10x10 mm Body [LQFP]
Atmel Legacy Global Package Code AKN

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



64-Lead Low Profile Quad Flat Pack (2CB) - 10x10 mm Body [LQFP] **Atmel Legacy Global Package Code AKN**

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Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	64		
Pitch	e	0.50 BSC		
Overall Height	A	-	-	1.60
Standoff	A1	0.05	0.02	0.15
Molded Plastic Height	A2	1.35	1.40	1.45
Overall Length	D	12.00 BSC		
Exposed Pad Length	D1	10.00 BSC		
Overall Width	E	12.00 BSC		
Exposed Pad Width	E1	10.00 BSC		
Terminal Width	b	0.17	0.22	0.27
Terminal Width	c	0.09	-	0.20
Terminal Length	L	0.45	0.60	0.75
Footprint	L1	1.00 REF		
Terminal Bend Radius	R1	0.08	-	-
Terminal Bend Radius	R2	0.08	-	0.20
Terminal Angle	Θ	0°	3.5°	7°
Terminal Angle	Θ1	0°	-	-
Mold Draft Angle	Θ2	11°	12°	13°

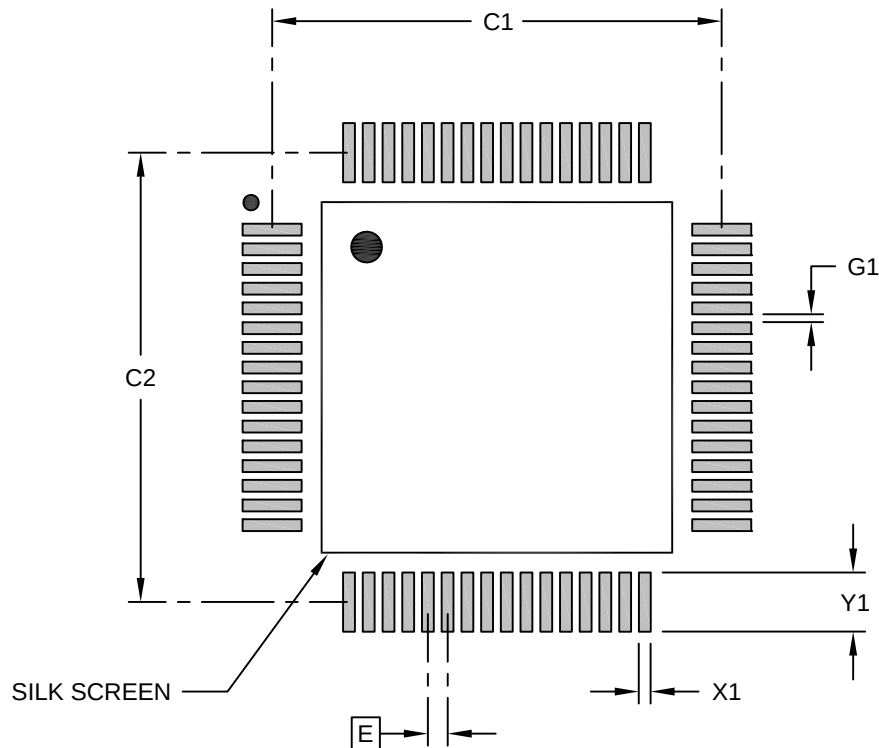
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

64-Lead Low Profile Quad Flat Pack (2CB) - 10x10 mm Body [LQFP]

Atmel Legacy Global Package Code AKN

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

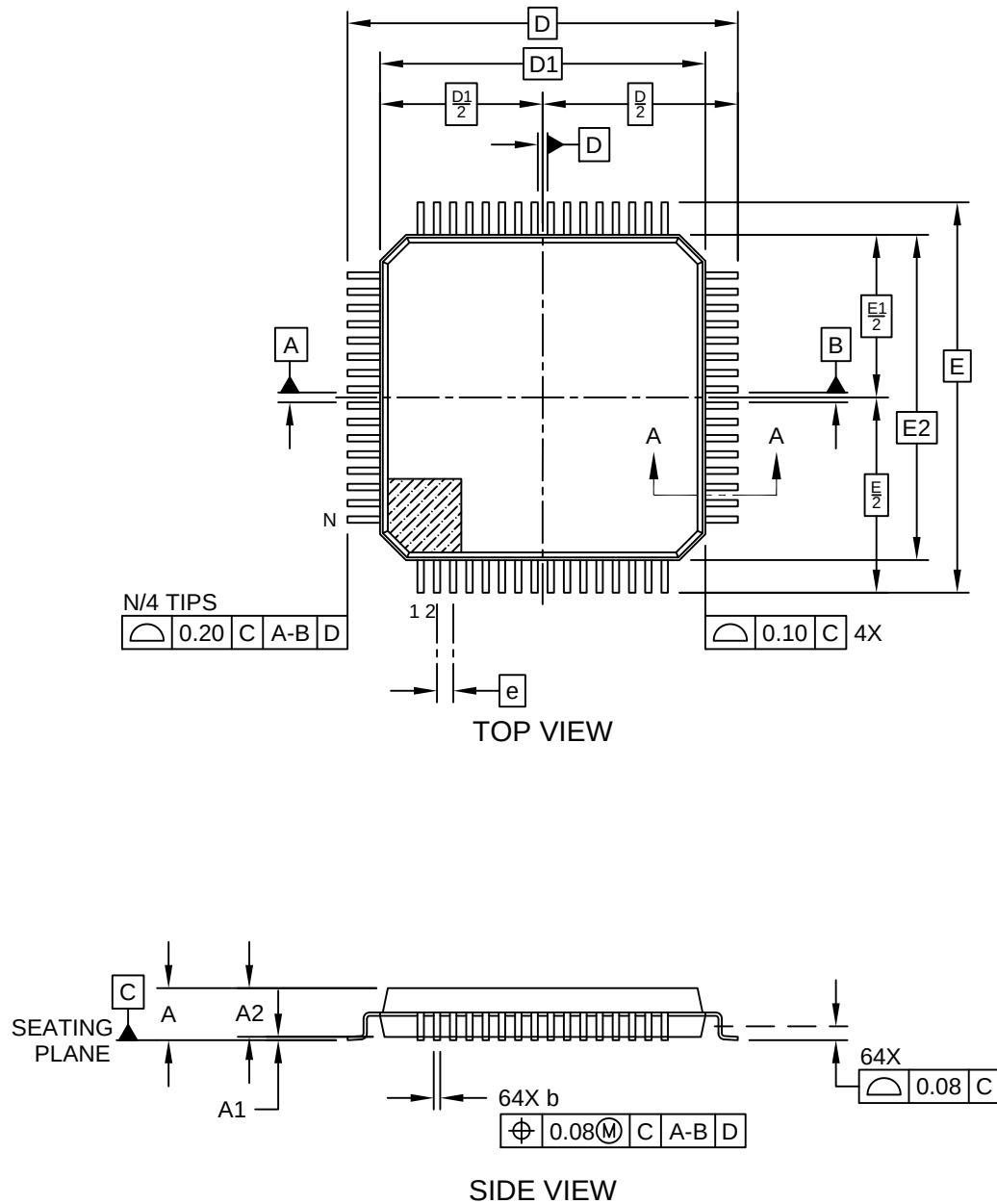
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Contact Pad Spacing	C1		11.40	
Contact Pad Spacing	C2		11.40	
Contact Pad Width (X64)	X1			0.30
Contact Pad Length (X64)	Y1			1.50
Contact Pad to Contact Pad (X60)	G1	0.20		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.

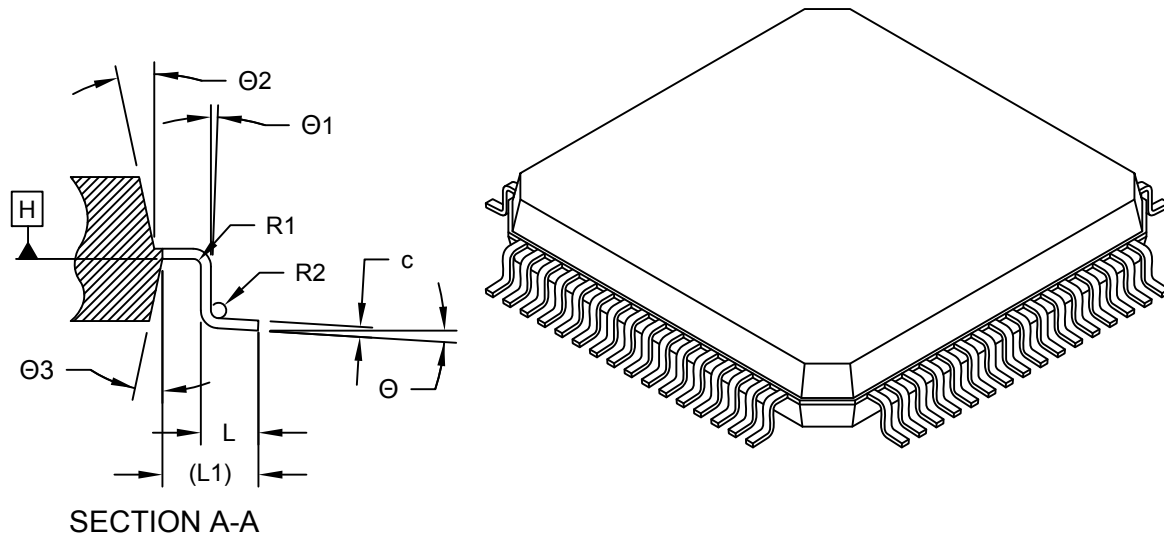
64-Lead Low Profile Quad Flat Pack (2DB) - 10x10 mm Body [LQFP]
Atmel Legacy Global Package Code AKN

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



64-Lead Low Profile Quad Flat Pack (2DB) - 10x10 mm Body [LQFP] **Atmel Legacy Global Package Code AKN**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



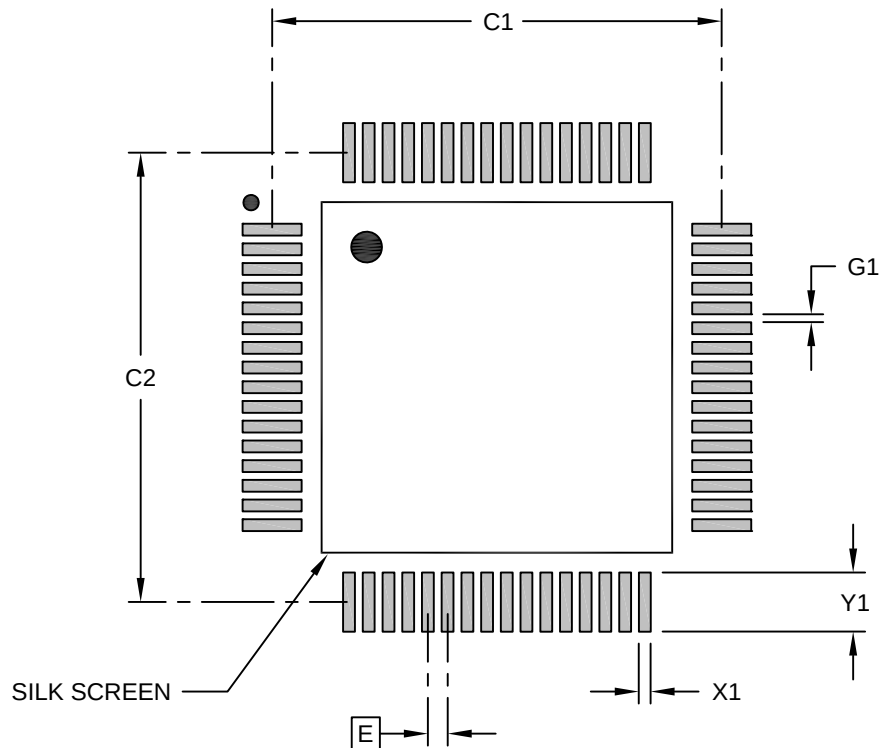
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	64		
Pitch	e	0.50 BSC		
Overall Height	A	-	-	1.60
Standoff	A1	0.05	0.02	0.15
Molded Plastic Height	A2	1.35	1.40	1.45
Overall Length	D	12.00 BSC		
Exposed Pad Length	D1	10.00 BSC		
Overall Width	E	12.00 BSC		
Exposed Pad Width	E1	10.00 BSC		
Terminal Width	b	0.17	0.22	0.27
Terminal Width	c	0.09	-	0.20
Terminal Length	L	0.45	0.60	0.75
Footprint	L1	1.00 REF		
Terminal Bend Radius	R1	0.08	-	-
Terminal Bend Radius	R2	0.08	-	0.20
Terminal Angle	Θ	0°	3.5°	7°
Terminal Angle	Θ1	0°	-	-
Mold Draft Angle	Θ2	11°	12°	13°

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

64-Lead Low Profile Quad Flat Pack (2DB) - 10x10 mm Body [LQFP] Atmel Legacy Global Package Code AKN

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Contact Pad Spacing	C1		11.40	
Contact Pad Spacing	C2		11.40	
Contact Pad Width (X64)	X1			0.30
Contact Pad Length (X64)	Y1			1.50
Contact Pad to Contact Pad (X60)	G1	0.20		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.